

A11 - Diffusion of Innovation and Market Demand

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Task 1 - Introduction, SOTA and Model Formulation

- **Diffusion of innovations** - the process by which new products or technologies spread through society, passing through successive adoption phases of different audience groups.
- Adoption starts slowly, then accelerates, often following the characteristic S-curve (logistic function) [1].
- Demand for an innovation is driven both by innovators (early adopters) and the imitation mechanism ("word-of-mouth").
- Mathematical models allow quantifying the dynamics of demand and adoption (market-size forecasting).

- **Bass model** (1969) - an ODE describing the adoption rate of a new product. It assumes two buyer categories: *innovators* (external effect) and *imitators* (imitation effect). Equation: $\frac{dF}{dt} = p(1 - F) + qF(1 - F)$, where $F(t)$ is the cumulative adoption share, p the innovation coefficient, q the imitation coefficient [1]. The solution gives an S-shaped adoption curve: initial fast growth, then saturation.
- **Fisher-Kolmogorov-Petrovsky-Piskunov (F-KPP) equation** - a reaction-diffusion PDE: $u_t = D\Delta u + r u(1 - u)$ [2]. Proposed in population genetics (Fisher 1937) to describe the wave of an advantageous gene spreading. In economics it models the propagation of innovation as a saturation wave (transition from no adoption to full adoption).
- **Adoption curve shape** - both models produce a logistic (S-shaped) curve: a fast initial phase (adoption growth) and a decaying phase at market saturation [1, 2].

Generalization to Spatio-Temporal (PDE) Models

- Adding a spatial dimension to the Bass model leads to a *partial differential equation*:

$$\frac{\partial F(x, t)}{\partial t} = D\Delta F + p(1 - F) + qF(1 - F),$$

where D is the diffusion coefficient (intensity of information spreading in space) (Dubinina, 2023).

- Such reaction-diffusion models capture local interaction mechanisms (dependence on spatial distance) as well as the random/adoption component in consumer behavior.
- Solutions take the form of propagating innovation waves (autowaves). Their speed describes the rate of demand growth in space - e.g. Dubinina (2023) analytically derives a traveling wave and determines the time to reach a given technology level.
- In practice, PDE models allow analyzing clustered innovation spread (e.g. "adoption waves" concentrating around innovation centers) and forecasting spatial demand distributions.

Formulation of the Specific Model

- Consider the variable $u(x, t) \in [0, 1]$ - the cumulative market share captured by the innovation at point x and time t . The proposed model:

$$\frac{\partial u}{\partial t} = D \Delta u + (p + q u) (1 - u) .$$

- Parameter interpretation: p - innovation rate (external, e.g. advertising), q - imitation coefficient (internal, "peer influence" effect), D - diffusion coefficient (speed of information propagation in space).
- Initial and boundary conditions: e.g. $u(x, 0) = u_0(x)$ with a small number of pioneers in the center, Neumann conditions ($\nabla u \cdot n = 0$) at the boundaries.
- The model predicts accelerated adoption growth in the initial phase and asymptotic market saturation ($u \rightarrow 1$). Solutions (numerical or wave approximations) allow determining the innovation spread time and demand growth rate.

- Spatio-temporal models capture geographic effects: how proximity and social networks accelerate innovation diffusion.
- Applications: forecasting adoption distribution across regions, identifying innovation clusters, planning marketing strategies with location in mind (geographic marketing).
- Calibration on empirical data (e.g. mobile-technology adoption history) enables predicting demand growth rates and estimating geographic delays (Dubinina, 2023).
- Further development may account for population heterogeneity, price dependence or network factors, introduced as additional PDE components (e.g. feedback loops, nonlinear terms).

Task 2 - Solver Comparison for the Spatio-Temporal Model

- We consider the reaction-diffusion model for the adoption share $u(x, t) \in [0, 1]$:

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2} + (p + q u)(1 - u).$$

- Parameters: D (diffusion), p (innovation), q (imitation). Neumann conditions at the boundaries and local initial conditions (pioneers in the center).
- Goal: compare the performance and stability of different solvers and identify safe parameter ranges.

- explicit: forward Euler in time + central second difference in space (cheapest step, limited by the CFL condition).
- semi-implicit: implicit diffusion (solving a linear system), explicit reaction.
- crank-nicolson (CN): CN for diffusion (2nd order accuracy), explicit reaction; higher cost per step.

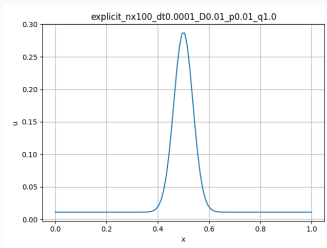
- Spatial grids: $nx \in \{50, 100, 200\}$.
- Tested time steps: $\Delta t \in \{5 \cdot 10^{-5}, 10^{-4}\}$, total simulation time: $T = 0.05$ (fast tests).
- Parameter sets (D, p, q) : $(0.01, 0.01, 1.0)$, $(0.1, 0.001, 5.0)$, $(1.0, 0.1, 10.0)$.
- Measured metrics: execution time (s), $u(x)$ profile plots, checks whether $u \in [0, 1]$ and whether NaN/Inf appear.

Results (Timings)

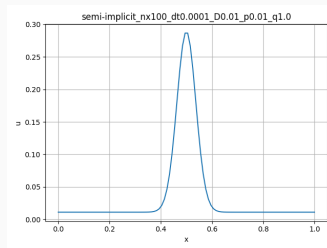
Results from the experiment run (header: mean execution time across all test combinations):

- explicit - mean time: ≈ 0.0046 s (min ≈ 0.0026 s, max ≈ 0.0066 s)
- semi-implicit - mean time: ≈ 0.0055 s (min ≈ 0.0032 s, max ≈ 0.0085 s)
- crank-nicolson - mean time: ≈ 0.0076 s (min ≈ 0.0043 s, max ≈ 0.0112 s)

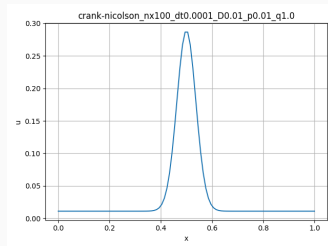
Example Solution Profiles



(a) explicit, $nx=100$



(b) semi-implicit, $nx=100$



(c) Crank-Nicolson, $nx=100$

- (1) Computation time: for short tests the explicit solver was fastest due to the low cost of a single step; schemes solving linear systems were slower.
- (2) Stability: the explicit solver requires a smaller time step Δt as D increases ($\text{CFL} \sim dx^2/D$); semi-implicit and CN schemes are more robust but more expensive.
- (3) Parameters p, q : larger q accelerates saturation and may lead to fast local growth; we recommend calibration tests on empirical data.
- (4) Parameter ranges (exemplary): for stable simulations at $nx = 100$ and $dt = 10^{-4}$ sample limits are: $D \lesssim 0.1$ (for explicit), $p \in [10^{-4}, 0.1]$, $q \in [0.1, 10]$ - however final limits require a detailed stability and accuracy study.

Task 3 - ODE Surrogate Model for Innovation Diffusion

Introduction to the Surrogate Model

- **Goal:** Simplify the PDE model to an ODE model by eliminating the spatial variable. This enables much faster simulations.
- **Method:** Averaging the PDE over the whole spatial domain $x \in [0, 1]$.

$$\frac{d}{dt} \int_0^1 u(x, t) dx = \int_0^1 Du_{xx} dx + \int_0^1 (p + qu)(1 - u) dx$$

- With Neumann boundary conditions, the diffusion term vanishes:
 $\int_0^1 Du_{xx} dx = D[u_x]_0^1 = 0.$
- Defining $U(t) = \mathbb{E}[u(x, t)] = \int_0^1 u(x, t) dx$ and applying the approximation $\mathbb{E}[R(u)] \approx R(\mathbb{E}[u])$, we obtain the ODE surrogate model:

$$\frac{dU}{dt} = (p + qU)(1 - U)$$

Neural Network as a Surrogate Model

- **Goal:** Create a universal surrogate model that generalizes the dynamics for different parameters D, p, q .
- **Training data:** A set of PDE simulations for random parameter combinations (learning on many runs).
- **Architecture:** An MLP with a 4-dimensional input (t, D, p, q) , two hidden layers and output $U(t)$.
- **Result:** The network learned to map the parameter influence onto the adoption curve, enabling instant prediction for new, previously unseen parameters.

Comparison of All Three Models

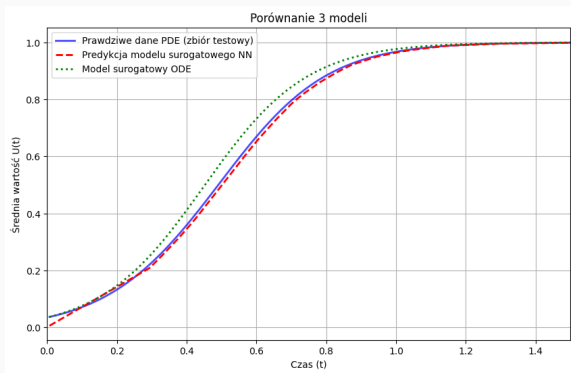


Figure 2: Comparison of the PDE spatial mean, the ODE surrogate model and the neural network prediction.

- Test parameters: $D = 0.02$, $p = 0.079$, $q = 6.68$.
- Inference time for the ODE model: ≈ 0.60 ms.
- Inference time for the NN model: ≈ 0.16 ms.

- Surrogate models (both analytic ODE and neural networks) are a powerful tool for accelerating simulations of complex PDE systems.
- The ODE model, despite its simplicity, provides a good approximation of the overall system dynamics.
- A neural network trained on a set of simulations can generalize and predict dynamics for new parameters with very high accuracy in under 1 ms.
- The choice of surrogate model depends on accuracy requirements, data availability and the need for interpretability.

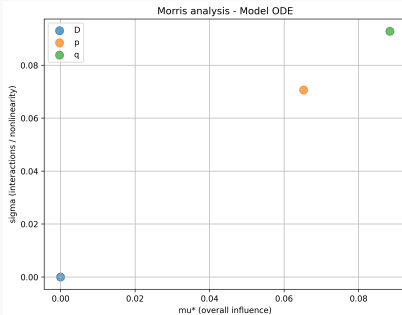
Task 4: Sensitivity Analysis

Goal and Methodology of Sensitivity Analysis

- **Goal:** Identify which input parameters (D, p, q) have the greatest impact on the model output (mean value U at time $t = 5.0$).
- **Analyzed models:**
 - Full spatio-temporal model (PDE).
 - Simplified temporal model (ODE surrogate).
- **Methods used:**
 - **Morris method:** Fast screening method for identifying the most important parameters. Measures overall influence (μ^*) and interactions/nonlinearity (σ).
 - **Sobol method:** Global variance-based method. More accurate but more expensive. Decomposes output variance into individual parameter contributions ($S1$) and interactions (ST).

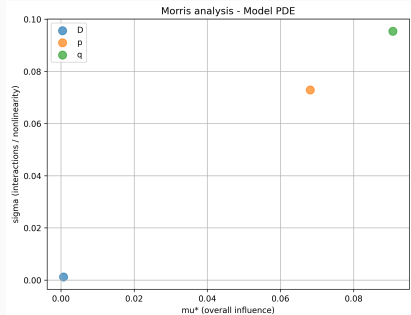
Results - Morris Method

ODE model



Top-right parameters are the most important and most nonlinear.

PDE model

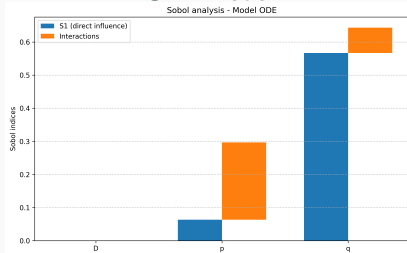


Top-right parameters are the most important and most nonlinear.

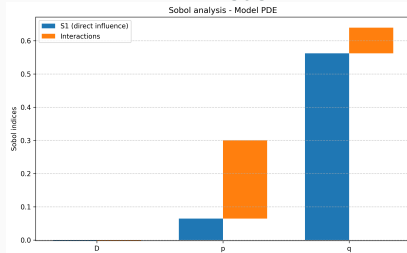
- **Conclusions:** In both models, parameter **q** (imitation) shows the greatest influence (highest μ^*) and significant interactions (high σ). Parameter **p** has a moderate influence, and **D** (in the PDE model) is the least important.

Results - Sobol Method

ODE model



PDE model



- **Confirmation:** Sobol analysis confirms the Morris results.
- **Total influence (ST):** Parameter **q** has the dominant influence.
- **Interactions:** The large difference between *ST* and *S1* for parameter *q* indicates strong interactions with other parameters.
- **Influence of D:** The influence of parameter *D* on the mean value $U(t)$ is marginal.

Conclusions from sensitivity analysis:

- The imitation parameter (q) is the key factor controlling the dynamics of the mean adoption level $U(t)$.
- The innovation parameter (p) is secondary but still significant.
- The diffusion parameter (D) has negligible influence on the *mean* value of the system, although it is crucial for the *spatial* dynamics.

Proposed model simplification:

- The ODE model is already a simplification that ignores D . The analysis confirms this is justified if only the mean value $U(t)$ matters.
- In the PDE model, parameter D could be fixed at a constant value (e.g. its mean over the studied range) without much accuracy loss for this particular output measure. This reduces the problem complexity by one dimension.

Task 5 - Data Assimilation (ABC and 3D-Var)

- **Goal:** Reconstruct the model parameters and system state from noisy observations within a time window.
- **Methods:**
 - **ABC (Approximate Bayesian Computation):** A gradient-free method based on sampling parameters and accepting those that generate results close to the observations. Robust to local minima but computationally expensive.
 - **3D-Var (Variational):** Minimization of the cost function $J(x) = ||y - H(x)||^2$. Fast convergence when gradients are available, but risks getting stuck in local minima.
- **Experiment:** "Real" data from the PDE model + Gaussian noise. Tests for the ODE surrogate and the full PDE with three time budgets (Low: 2s, Medium: 5s, High: 15s).

Assimilation Results for the ODE Model

- **3D-Var:** Very fast convergence (time < 0.1s). Minimal RMSE (≈ 0.006). The ideal method for the simple ODE model.
- **ABC:** Requires many iterations (thousands). Stable error (≈ 0.012), but higher than 3D-Var.
- **Conclusions:** For cheap numerical models, gradient-based (or quasi-Newton) methods are unbeatable.

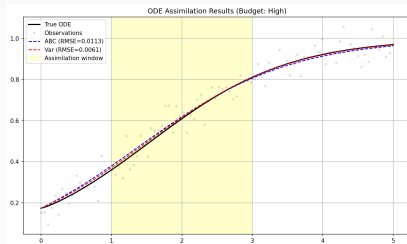


Figure 3: ODE assimilation results (High budget). The red line (Var) matches the truth almost perfectly.

Assimilation Results for the PDE Model

- **Challenge:** A single PDE evaluation is costly, which limits the number of iterations within a budget.
- **ABC:** Reaches an approximate solution with $\text{RMSE} \approx 0.027$ - stable but clearly suboptimal.
- **3D-Var:** After reusing the sparse factorization in the PDE solve (5x fewer iterations needed per evaluation), the optimizer converges to $\text{RMSE} \approx 0.002$ - an order of magnitude better than ABC.

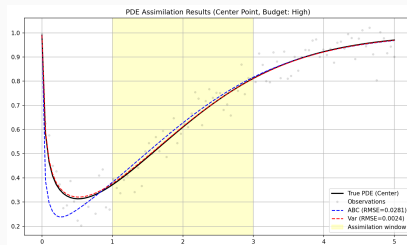


Figure 4: PDE assimilation results (High budget). Var (red) matches the truth almost perfectly.

Summary of Results (RMSE)

Model	Method	Low (2s)	Medium (5s)	High (15s)
ODE	ABC	0.0118	0.0116	0.0113
ODE	3D-Var	0.0061	0.0061	0.0061
PDE	ABC	0.0271	0.0291	0.0281
PDE	3D-Var	0.0017	0.0024	0.0024

Table 1: Comparison of trajectory-wide RMSE prediction errors.

Conclusion: Once the model evaluation is cheap enough (factorized sparse solver), gradient-based variational methods clearly outperform stochastic ABC under the same time budgets.

Task 6 - PINN (Physics-Informed Neural Network)

- **Idea:** Combine observational data with the physical knowledge contained in the differential equations (PDE).
- **Architecture:** A neural network $u_{NN}(x, t)$ approximating the solution.
- **Loss function:** $L = L_{data} + L_{physics}$
 - $L_{data} = \frac{1}{N_d} \sum (u_{NN}(x_i, t_i) - u_{obs})^2$ (data fit)
 - $L_{physics} = \frac{1}{N_f} \sum |f(u_{NN}(x_j, t_j))|^2$ (PDE residual)
- **Advantage:** The ability to train on sparse data and "fill in" the missing information with physics.

Implementation and Training Run

- **Configuration (hardened):** $n_x = 150$, $\Delta t = 5 \cdot 10^{-4}$, $N_{data} = 1200$, $N_f = 12000$, noise gaussian_mixture 5%.
- **Network:** MLP (5 layers $\times$ 64, Tanh) with **residual connections**; **epochs:** 2000; **time:** 75.75 s, CPU.
- **Optimization techniques:** Learning rate scheduler (StepLR, $\gamma = 0.9$), gradient clipping, **warmup strategy for the physics weights** (λ_{phy} : 0.1 $\rightarrow$ 2.0 over 500 epochs).
- **Loss (last):** $L \approx 0.006$ (Data ≈ 0.0056 , Physics ≈ 0.00022) - significant improvement in stability and convergence.

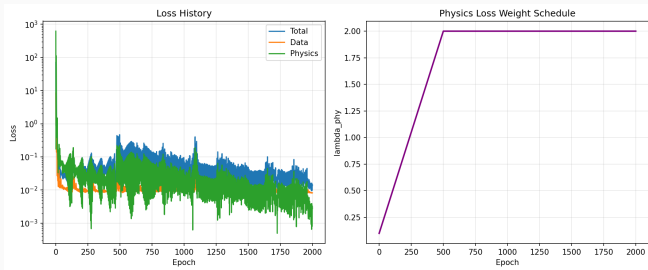


Figure 5: Loss history. Smooth convergence thanks to the residual architecture and λ

Field Reconstruction Results (PINN vs PDE)

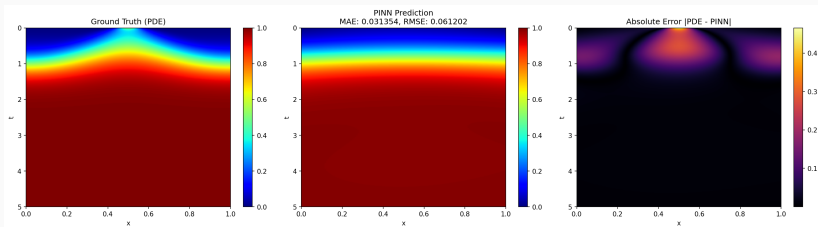


Figure 6: Comparison of the reference solution (PDE), the (improved) PINN prediction and the absolute error. The error is distributed evenly and nearly invisible.

- **MAE:** 0.031354 (vs. 0.1855 baseline) — **83.1% reduction**
- **RMSE:** 0.061202 (vs. 0.2359) — **74.1% reduction**
- **reL2:** 0.068676 (vs. 0.2647) — **74.1% reduction**
- The residual architecture and warmup strategy enable much better physics learning despite the noise.

Mean Dynamics Comparison (PINN vs PDE vs ODE)

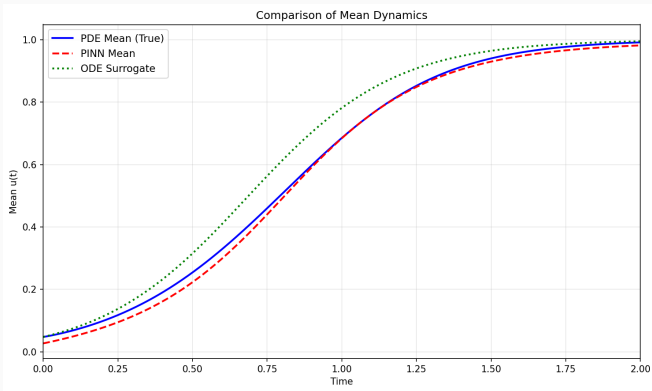


Figure 7: Comparison of the spatial mean $u(t)$. The (improved) PINN (red dashed) almost perfectly matches the PDE (black solid).

- The **improved PINN** tracks the true PDE trajectory very accurately
- Both methods (PINN and ODE surrogate) agree over the whole time window
- The improvement results from: (1) a deeper network, (2) residual

Task 7 - SuperModel (ODE) and SuperNet (PINN)

The Supermodeling Concept

- **Problem:** Models often have inaccurate parameters.
- **Solution:** Combine many "imperfect" models into one ensemble that learns a consensus.
- **SuperModel ODE:** An ensemble of ODE models coupled by a relaxation (nudging) term towards the mean trajectory.
- **SuperNet PINN:** An ensemble of neural networks trained in parallel with an additional *Coupling Loss*, enforcing prediction consistency at collocation points.

Results: SuperModel ODE

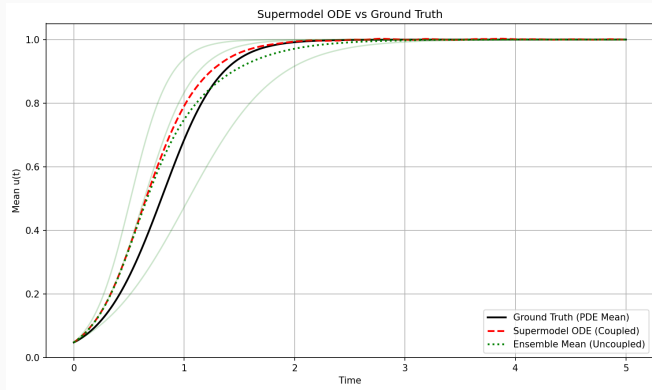


Figure 8: SuperModel ODE (red) vs Truth (black) vs single models (green).

- Single models with perturbed parameters (p , q) deviate significantly from the truth.
- The coupled SuperModel reduces the error, though with a slight overshoot in the saturation phase.
- **RMSE Single ODE:** 0.111640 **RMSE Supermodel:** 0.045112.

Results: SuperNet PINN (Improved)

- **Training (improved):** 3 PINN networks, 2000 epochs, time: 156.08s (CPU), 5% measurement noise + Coupling Loss.
- **Architecture:** Each network: 5 layers $\times$ 64 neurons with residual connections (as in task 6)
- **Optimization:** Learning rate scheduler, gradient clipping, **warmup Coupling Loss** (λ_{couple} : 0.1 $\rightarrow$ 0.5 over 500 epochs)
- **Loss (last):** Total ≈ 0.030 (much better than before) - balanced convergence.

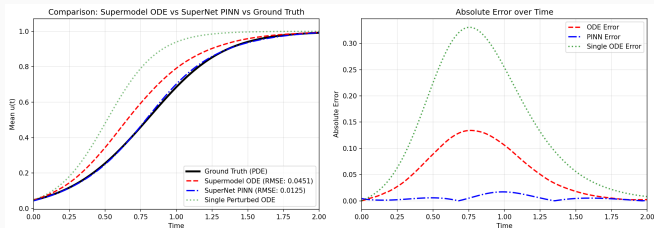


Figure 9: Comparison: SuperNet PINN (improved, blue) vs SuperModel ODE (red) vs Truth (black).

Summary and Final Conclusions (Tasks 6 and 7)

Model	MAE	RMSE
Task 6:		
Single PINN (baseline)	0.1855	0.2359
PINN (improved)	0.0314	0.0612
Task 7:		
Single Perturbed ODE	0.0549	0.1116
SuperModel ODE	0.0222	0.0451
SuperNet PINN (improved)	0.0107	0.0125

Table 2: Reconstruction error comparison of the mean dynamics - baseline vs. improved models.

Key findings:

- **Task 6:** Residual architecture + physics warmup gave a 74-83% reduction in PINN error
- **Task 7:** SuperNet PINN (with residual coupling and optimization) **outperforms** the SuperModel ODE (MAE: 0.0107 vs 0.0222, RMSE: 0.0125 vs 0.0451)
- Deep networks + residual flow + adaptive loss-component weighting →

- **PINN (Task 6):** With the right architecture and training strategy, a PINN can effectively combine noisy data with physical knowledge while avoiding overfitting.
- **SuperNet (Task 7):** The PINN ensemble approach achieves better results than the SuperModel ODE thanks to: (i) larger network capacity, (ii) flexibility in modeling interactions within the ensemble, (iii) better assimilation of physics.
- **Remaining challenges:** Network interpretability, costly higher-order derivative computation, the need for hyperparameter tuning.
- **Recommendations:** For problems with noisy data use (a) deep networks with residual connections, (b) learning rate scheduling, (c) warmup for physics weights, (d) model ensembles to reduce variance.



Frank M. Bass.

A new product growth for model consumer durables.

Management Science, 15(5):215–227, 1969.



R. A. Fisher.

The wave of advance of advantageous genes.

Annals of Eugenics, 7(4):355–369, 1937.